

A Stratified Analysis of Link Prediction Fairness

Weixiang Wang and Sucheta Soundarajan

Syracuse University, Syracuse NY 13244, USA
{[wwang69](mailto:wwang69@syr.edu), [susounda](mailto:susounda@syr.edu)}@syr.edu

Abstract. Link prediction systems are widely used in many domains, and fairness has increasingly become a focal point of research. Existing studies on link prediction fairness aggregate results within protected groups, losing fine-grained topological information and ignoring the potential effects of Simpson’s Paradox. In this work, we propose a fine-grained fairness metric, *Stratified Disparity*, which uses a layered analysis to capture unfairness between different node groups. Our experiments, applying *Stratified Disparity* to several real-world datasets and link prediction algorithms, demonstrate the existence of Simpson’s Paradox, suggesting that traditional fairness evaluation methods can indeed mask or amplify unfairness for certain groups, failing to reflect individual-level disparities and potentially leading to misjudgments of unfairness.

Keywords: fairness, link prediction, Simpson’s Paradox

1 Introduction

Link prediction algorithms are widely used across various domains, including for friendship recommendations in social networks, collaboration predictions in academic networks, and citation predictions in research networks. Recently, concerns regarding fairness in link prediction models have gained significant attention [5, 6, 10]. In algorithmic settings, fairness is usually defined with respect to performance disparities between various protected groups of interest (based, e.g., on race or gender). However, the analysis of algorithmic fairness originated primarily in the non-relational machine learning community, and while many techniques are generalizable, a network setting poses distinct and important challenges.

Our goal in this work is to propose a fundamentally new approach to analyzing network algorithmic fairness, one in which network topology, rather than just node attributes, takes a central role, and the potential presence of Simpson’s Paradox¹ is fully acknowledged.

In non-relational/non-network machine learning contexts, unfairness is often due to correlations and relationships between attributes: for example, if historical data exhibited a bias against a protected group (such as women in medical school admissions), then models trained on that data may perpetuate the bias. In a network context, unfairness may additionally arise because of topological

¹ Simpson’s Paradox refers to a phenomenon in which a property that appears when data points are aggregated disappears or reverses when they are subdivided.

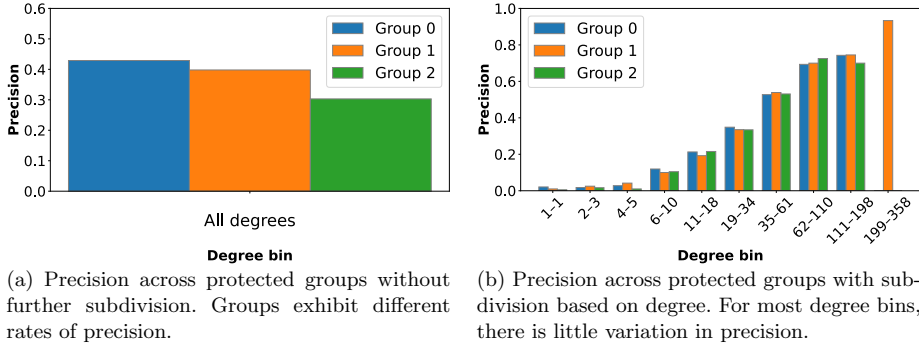


Fig. 1. Node-level precision statistics for the **Bowdoin** dataset. For each node u , Jaccard index is used to identify the 10 highest-scoring nodes in u 's 2-hop neighborhood.

differences between protected groups. For instance, if one protected group occupies higher-density sections of a network, there may be more and richer data from which inferences can be made about members of that group.

Disentangling the role of network topology vs. node attributes in causing algorithmic unfairness is crucial in developing a solution to such unfairness. If caused by differences in attributes, then one might address unfairness by certain types of pre-processing, including modification of attributes or embeddings, possibly adapted from the traditional machine learning setting. In contrast, if the issue is that certain protected groups simply do not have sufficient data, then the problem is much harder to address. Although we focus on link prediction, the broad approach can be generalized to other analyses of network fairness.

We propose a metric called *Stratified Disparity* which, in addition to comparing performance differences between protected groups, also considers the effect of topological properties. In this work, to demonstrate the metric, we consider degree, though the method can easily be applied to other properties or combinations of properties. Our analysis finds that much observed unfairness arises from differences in group degree, rather than differences in protected attribute.

As an example, see Figure 1, which shows results on the **Facebook100 Bowdoin** [15] network (G) based on the **gender** attribute. For this analysis, we treat the edges that do not exist in G within a node's 2-hop neighborhood as candidate edges for recommendation. (This neighborhood was chosen for efficiency reasons; but a user can easily use whatever search space they wish.) We then recommend the top-10 edges with the highest Jaccard similarity scores [8] to each node and calculate the prediction precision for each node. Precision differences across protected groups both without (Figure 1(a)) and with (Figure 1(b)) degree stratification are shown. We observe a standard deviation of 0.0657 across group-level average precisions when nodes are not stratified by degree, but an average per-bin standard deviation of only 0.0099 across group-level average precisions when they are stratified, strongly suggesting that degree differences are primarily responsible for performance disparities.

Importantly, this observation suggests that existing approaches to network fairness are insufficient. Many “fairness-aware” link prediction algorithms are predicated on the assumption that differences in group performance are best explained by attributes, and if such differences are actually caused by topological differences, then a new approach is required.

We evaluate the proposed *Stratified Disparity* on eight popular link prediction algorithms across eight real-world datasets. We demonstrate that comparing precisions of protected groups directly against one another can mask or amplify unfairness, but when groups are subdivided by degree, differences become more apparent, or vanish. These results have ramifications for both the evaluation processes and design of link prediction algorithms.

2 Background

Link Prediction: Link prediction algorithms are used in many real-world applications. For instance, in online social networks, link prediction is used to recommend potential friends to users. In the link prediction problem, one often assumes that there is an observed network that is a subset of a larger, unobserved network. The goal is to predict which edges that do not exist in the observed network are most likely to exist in the full network.

To make predictions, link prediction algorithms may use network topological features such as common neighbors, as well as vertex attributes (e.g., interests and beliefs), to make predictions. It is common that such algorithms score non-existent edges, and then suggest them in decreasing order of score. Some of the oldest, but still highly relevant, methods include those based on Jaccard similarity, random walks, and common neighbors [8]. More recent methods use machine learning to combine features into a model.

Algorithmic Fairness: Algorithmic fairness measures are typically used to evaluate differences in algorithm performance across protected groups, such as those representing different genders, nationalities, and so on. One common metric is the *statistical parity* metric, which, when applied to link prediction, requires that the probability that an edge (u, v) is predicted should be the same regardless of whether nodes u and v are in the same group or in different groups [4]. In the *equality of odds* metric, fairness is measured as the link prediction accuracy difference between inter-group and intra-group edges [4]. Such disparities may, for example, hinder cross-group interactions and reinforce structural segregation in networks. One limitation of the above metrics is that they aggregate nodes from the same group to compute fairness without considering their distribution, which can lead to biased measurements.

Link prediction unfairness can arise from multiple sources, with data being a critical factor. Data imbalance often leads to algorithmic bias towards certain groups. Even with balanced data, biases can still be introduced during algorithm design and optimization. Certain models may overly rely on protected attributes (e.g., gender, race). When addressing fairness in link prediction, the goal is to ensure that the model does not perpetuate or amplify inherent biases present in

the network data, and that the predicted results are fair across different groups or communities within the network. This is particularly crucial in contexts where network data reflect historical inequalities, discrimination, or social biases.

Fairness-aware link prediction methods can be broadly grouped into three categories. The first category focuses on learning fair node representations, and includes GFairHint [18], FLIP [10], FairWalk [4], and CrossWalk [4], which generate debiased embeddings for downstream link prediction. The second category improves fairness through graph preprocessing, with methods such as FairLP [7], EDITS [3], UGE [16], and FairLink [9]. The third category, including methods such as FairLink [17], adopts reinforcement learning-based approaches which optimize fairness-related rewards to guide edge addition.

The relationship between Simpson’s Paradox and fairness is known: for example, Prost et al. [12] explored Simpson’s Paradox in recommender fairness. However, discussions of Simpson’s Paradox in link prediction are absent.

3 Stratified Disparity

Unfairness in network algorithms can be caused by features, topology, or both, and understanding the cause of such unfairness is important in determining a strategy to reduce that unfairness.

Before defining the proposed *Stratified Disparity* metric, consider again the example in Figure 1, which shows the **Bowdoin network**, an early Facebook network with the gender attribute (male, female, and unknown). For each node, a link prediction algorithm recommends the top-10 edges with the highest Jaccard similarity scores from its 2-hop neighborhood (if fewer than 10 candidates satisfy, we recommend only those edges), and from this, we calculate the **prediction precision** for each node. The choice to restrict to the 2-hop neighborhood is made for the sake of efficiency, but is not a fundamental restriction of the approach.

When nodes are not separated by degree (Figure 1(a)), the average precisions across protected groups are 0.429, 0.398, and 0.303, respectively. These values are the basis of a traditional fairness comparison: comparing precisions across groups, without further subdivision. In contrast, if we stratify by degree (h (Figure 1(b)), we see that the standard deviation between protected group precisions within each degree-bin are, on average, 84.9% lower: from 0.0657 without stratification to 0.0099 within 10 stratified bins, prior to accounting for sampling noise. This motivates the main intuition behind *Stratified Disparity*, which is to account for topological differences when measuring unfairness.

We base the proposed *Stratified Disparity* metric on results such as these. *Stratified Disparity* does more than compute the change in standard deviation: it additionally controls for sample noise and imputes missing values. On **Bowdoin network**, *Stratified Disparity* reports a reduction of approximately 25% from the unstratified case to the stratified case, indicating that a significant portion of the unfairness can be explained by differences in degree.

3.1 Stratified Disparity: Definition

To compute STRATIFIED DISPARITY, a user provides:

1. Link prediction precisions for each node in that network.
2. For each node, a categorical protected attribute $A_i \in \{1, \dots, t\}$ (e.g., gender).
3. For each node, a topological ‘binning’ attribute (e.g., degree) used to stratify across protected groups.
4. A difference function $\text{Diff}(g_1, \dots, g_t) \geq 0$ that computes the performance difference for each bin, where g_i represents the average precision of i th group in that bin. (More details are provided in Section 3.2).

Stratified Disparity measures the expected difference in link prediction precision across groups defined by the protected attribute, stratified by the binning attribute. First, we define *StratifiedDisparity*(n), which measures the Stratified Disparity for a given number of bins n , as:

$$\text{StratifiedDisparity}(n) = \sum_{j=1}^n \text{Prob}_j \times \text{Diff}_j(g_1, \dots, g_t), \quad (1)$$

where Prob_j is the probability that a randomly selected node (or edge) belongs to bin j , $\text{Diff}_j(g_1, \dots, g_t)$ denotes the performance difference within the j th bin. This is the weighted average across bins of the differences between groups. When $n = 1$, i.e., *StratifiedDisparity*(1), represents the scenario where all nodes are analyzed within one bin, aligning with conventional fairness metrics.

By computing *StratifiedDisparity*(n) for all possible bin counts n from 1 to l , where l is the number of distinct binning attribute values that appear (e.g., the number of distinct node degrees), one can define a curve in which the x -axis represents n , and the y -axis represents *StratifiedDisparity*(n).

Finally, the overall *Stratified Disparity* is defined by the elbow point *StratifiedDisparity*(e) of this curve (see Figure 2 for an example). We use the elbow point rather than the minimum value because it offers a compromise between number of bins and value. (Keeping the number of bins low is desirable if one wishes to design a link prediction algorithm intended to optimize for *Stratified Disparity*, because computational cost is lower.) For a given dataset, a lower *StratifiedDisparity*(e) indicates better fairness performance. This value enables fair comparison across link prediction algorithms.

3.2 Notes on Implementation

When computing *Stratified Disparity*, the user must make a number of design decisions, such as the choice of difference function, binning strategy, and how to handle bins without representatives from some of the protected groups. We give brief guidance as to these choices, but due to space constraints, cannot provide an exhaustive discussion.

The user may design the difference function $\text{Diff}(g_1, \dots, g_t)$ in a number of ways. Simple choices include the maximum or average difference between group

precisions. In this work, we consider the standard deviation across the protected group precisions in a given bin. To mitigate such noise, we randomly shuffle the protected attribute values across nodes twenty times to estimate the expected standard deviation arising purely from random noise, and subtract this expected standard deviation from the observed disparity (standard deviation).

Another design choice a user must make is in the binning strategy. Because our datasets are characterized by a heavy-tailed degree distribution, we adopt a binning strategy that uses equal-interval-width bins in the $\log_{10}(\text{degree})$ space. In some cases, certain protected groups may be absent from some of the bins. In these cases, we use the **K-Nearest Neighbor Imputation Method** (KNN imputer) [2] with $k = 2$, to estimate missing values, using the k bins adjacent to the missing entry in the sequence ordered by increasing bin counts.

3.3 Analysis of Stratified Disparity Curve

To gain additional insight into fairness, one can examine the disparity values as n varies. Increasing values of n produce a curve such as that shown in Figure 2, which provides an example for the **Bowdoin** network.

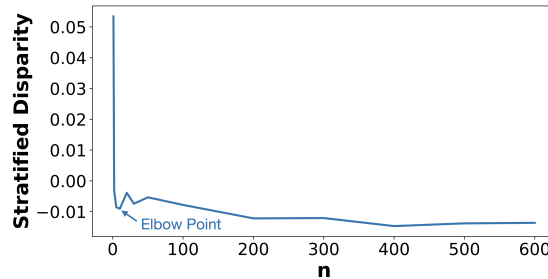


Fig. 2. Stratified Disparity Curve of Jaccard similarity on Bowdoin network as n increases, with missing values completed using k -nearest neighbors imputation.

The trend of the disparity curve gives important insights into link prediction fairness. As n increases, both increases and decreases in disparity highlight different flaws in conventional fairness measurement due to uneven distributions.

When analyzing disparity across multiple bins, the Stratified Disparity Curve typically demonstrates one of three broad trends: the curve may remain stable, increase, or decrease, each reflecting different types of algorithmic behavior. First, the Stratified Disparity may remain largely unchanged across bin count, indicating that aggregate and stratified disparities lead to consistent conclusions and that structural factors do not substantially affect the fairness assessment. Second, the Stratified Disparity may increase, suggesting that aggregate metrics obscure subgroup-level performance differences and that the algorithm behaves differently across structural contexts. Third, the Stratified Disparity may decrease, indicating that the aggregate disparity is driven primarily by differences in the structural distribution of the groups rather than by true performance differences within groups.

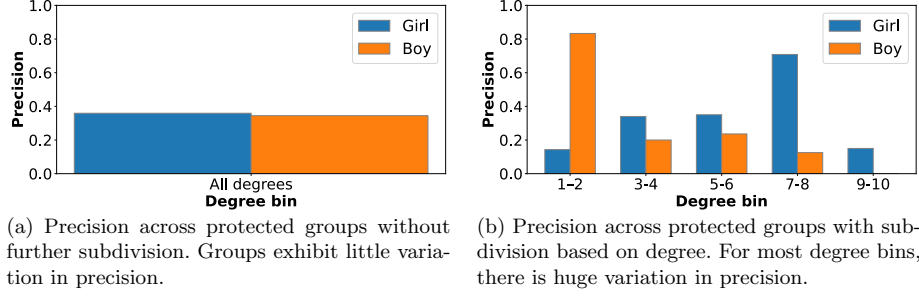


Fig. 3. Node-level precision statistics for the **Dutch School** dataset, using **Jaccard** similarity to predict edges.

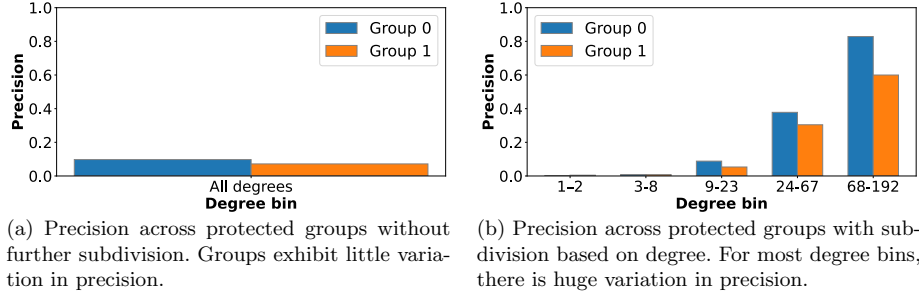


Fig. 4. Node-level precision statistics for the **Polblogs** dataset. For each node u , Jaccard index is used to identify the 10 highest-scoring nodes in u 's 2-hop neighborhood.

This nuance is particularly important when designing and evaluating a fairness-aware link prediction algorithm. With too coarse a view of fairness, it is possible that an algorithm attempts to reduce the precision gap between protected groups by unfairly favoring the disadvantaged group in a very small number of special bins. Fairness issues could thus be exacerbated without being easily detected. For example, to amplify the voices of minority groups on social media, the algorithm might promote already successful individuals who represent a tiny fraction of the group, while overlooking the voices of the group's marginalized members.

Next, to illustrate the advantage of Stratified Disparity and demonstrate how one might identify a case of Simpson's Paradox, we briefly jump ahead to analysis on real-world data. First, using the previous example of the **Bowdoin Network**, we observe that the disparity decreases from $StratifiedDisparity(1) = 5.34\%$ to $StratifiedDisparity(10) = -0.91\%$. This occurs primarily because group 1 contains more high-degree nodes, leading to a higher overall precision for this group. However, when comparing the performance within each bin, there is no meaningful difference: a classic example of Simpson's Paradox. In contrast, when analyzing the **Dutch School Network** (Figure 3), the disparity with one bin is $StratifiedDisparity(1) = 1.0\%$, while the disparity for five bins is $StratifiedDisparity(5) = 13.1\%$ — a 13-fold increase. In Dutch School Network, the Jaccard algorithm achieves higher accuracy to low-degree boys and high-

degree girls. Despite this disparity, aggregate unfairness is low: here, traditional metrics mask unfairness. Results on the **Polblogs Network** (Figure 4) show that the corresponding disparities are $\text{StratifiedDisparity}(1) = 1.37\%$ and $\text{StratifiedDisparity}(5) = 1.53\%$. Due to the presence of a power-law distribution, the precision in the first bin almost completely determines the overall precision. Consequently, although there is a significant precision disparity in subsequent bins, it is not reflected in traditional metrics.

4 Experiments

In this section, we discuss our network datasets and experimental setup. We apply our metric to eight baseline algorithms on eight real datasets.

Datasets: We consider the following datasets, which include friendship networks, co-authorship networks, and citation networks. Protected node attributes include gender, language and political party. In our datasets, the protected attributes are all specified as part of the data. Dataset statistics are shown in Table 1.

The **Dutch School Network** [10] dataset represents the friendship relations among 26 freshmen at a secondary school in the Netherlands and measures fairness with respect to the ‘Gender’ attribute.² The **Bowdoin** [15] network, from the Facebook100 dataset collection, represents a college social network, and we measure fairness with respect to the Gender attribute. The **Data Mining(DM)** and **Graphics** [14] are subsets of the DBLP network corresponding to those two subfields, for papers published between 2015 and 2019.³ For each author, we use gender as the protected attribute, where gender is obtained from the genderize.io tool. The **Pokec** [13] dataset represents an online social network in the town of Bytča in the Zilinsky Kraj region of Slovakia, where language is the protected attribute (German/English vs. non-German/English). The **Polblogs** [1] dataset represents hyperlinks between political blogs, and the protected attribute is whether a blog is liberal/conservative. The **UNC** [4] dataset is a social network from the University of North Carolina at Chapel Hill, with gender as the protected attribute. The **Recidivism(Recid.)** [4] dataset contains nodes representing defendants released on bail in U.S. state courts between 1990 and 2009, where edges represent similarity. Race is the protected attribute.

Link Prediction Algorithms: For each node u , we treat the set of edges that do not exist between u and nodes within its 2-hop neighborhood as candidate set $C(u)$. The algorithm outputs a score $s(u, v)$ for each $(u, v) \in C(u)$. We define $P(u)$ as the predicted-positive edges set of an algorithm. We then recommend the top $\min(10, |P(u)|)$ edges in $P(u)$ ranked by $s(u, v)$, and then compute the prediction precision for each node.

The first two methods we consider focus solely on accuracy. **Jaccard** sorts candidate edges based on the Jaccard similarity between the neighborhoods

² Due to the extremely small network size, we partition nodes using equal-width bins in the original degree space, rather than in the $\log_{10}(\text{degree})$ space.

³ <https://webdocs.cs.ualberta.ca/~zaiane/htmldocs/ConfRanking.html>

Table 1. Dataset statistics

Name	Dutch	Bowdoin	Pokey	DM	Graphics	Polblogs	UNC	Recid.
# Nodes	26	2,252	10,601	2,272	3,525	1,224	4,018	18,876
# Edges	221	84,387	127,594	7,643	10,399	16,718	65,287	330,746
Homophily	0.294	0.02	0.01	0.05	0.07	0.41	0.04	0.04
Intra Density	0.3198	0.0184	0.0024	0.0018	0.0010	0.0184	0.0041	0.0009
Inter Density	0.0523	0.0217	0.0019	0.0032	0.0019	0.0022	0.0035	0.0008
Prot. Attribute	gender	gender	language	gender	gender	pol. party	gender	race
Max Degree	10	358	327	32	44	192	107	2,508
# Other Attrs	0	0	0	0	0	0	8	18

of their endpoints [8]. **DeepWalk** learns a 64-dimensional embedding for each node; and for each candidate edge, concatenates the embedding vector for each endpoint to obtain a 128-dimensional feature vector. A two-layer neural network model then uses this feature vector to predict whether the edge exists [11].

Next, we consider three methods that improve fairness by pre-processing the network. For each method, we pre-process the network, then apply the **Jaccard** algorithm. **FairLP** balances densities of between-group and within-group edges by inserting new edges into the network, using the **MaxD** policy [7]. **UGE** learns node embeddings from a graph modified to reduce the influence of the protected attribute [16].⁴ **EDITS** quantifies discrimination between various protected groups using the Wasserstein distance, and prunes the topology and adjusts node features to minimize inter-group disparity [3].⁵

Finally, we consider three methods that attempt to learn unbiased node embeddings. We use the same experimental configuration as DeepWalk combined with a two-layer neural network, but replace DeepWalk with the corresponding fairness-aware method. **FairWalk** is a fairness-aware version of DeepWalk, and uses protected attributes to ensure that the random walk is equally likely to visit each group [4]. **CrossWalk** is a random-walk-based approach that encourages cross-group exploration and assigns weights that account for multi-hop neighborhood structure [4].⁶ **FLIP** is a GAN-based framework that generates node representations that retain structural information while suppressing the ability to distinguish between inter- and intra-group edges [10].⁷

5 Results

Our intention in proposing Stratified Disparity as a metric was to disentangle the causes of unfairness in link prediction. Indeed, experimental results reveal a wealth of observations. Complete results, evaluated using the elbow-point Stratified Disparity metric, are presented in Table 2. These results provide a basis for selecting the method with the lowest Stratified Disparity.

⁴ <https://github.com/MyTHWN/UGE-Unbiased-Graph-Embedding>

⁵ <https://github.com/yushundong/EDITS>

⁶ <https://github.com/ahmadkhajehnejad/CrossWalk>

⁷ <https://github.com/farzmas/FLIP>

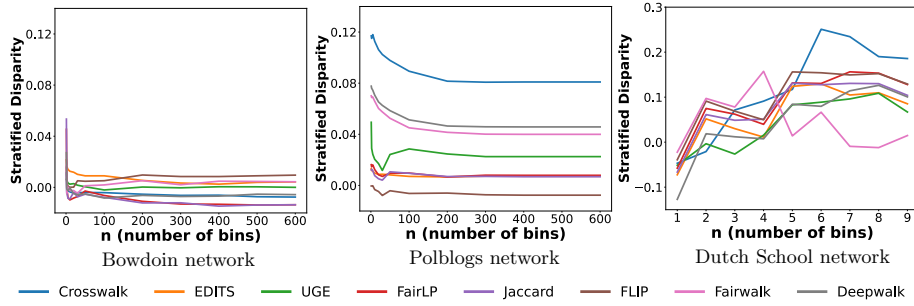


Fig. 5. Stratified Disparity Curves of existing link prediction algorithms on various networks, where lower values indicate smaller group-level performance disparities.

Network Comparison: A major takeaway from our results is that Simpson’s Paradox does indeed exist with traditional measures of link prediction fairness, and can both mask or amplify unfairness.

Figure 5 shows example results for eight link prediction methods on the Bowdoin, Polblogs and Dutch School networks. On the **Bowdoin** network, Stratified Disparity rapidly decreases toward zero as the number of degree bins increases. This behavior indicates that the observed aggregate disparity is largely attributable to degree-related effects. In contrast, the **Polblogs** network exhibits relatively stable disparity across stratification levels. This stability implies that degree is not the dominant factor driving performance differences across protected groups. Instead, the remaining disparity is likely induced by attribute alignment or other structural properties beyond degree. The **Dutch School network** demonstrates the opposite pattern. As the level of stratification increases, Stratified Disparity rises sharply, revealing substantial hidden bias that is masked when degree is ignored. Consequently, the algorithms exhibit significantly stronger bias with respect to the protected attribute than suggested by unstratified evaluation.

Stratified Disparity reveals that the algorithm exhibits opposite advantages across different structural strata: in some subsets it favors nodes from girls, while in others it favors nodes from boys. When results are aggregated, these opposing effects partially cancel each other out, masking the underlying unfairness. By evaluating disparities within strata, Stratified Disparity exposes these hidden biases that remain invisible under aggregate metrics.

Algorithm Comparison: Similarly, this analysis gives insights into the performance of different link prediction algorithms. When comparing algorithms, the main takeaway is that fairness-focused link prediction algorithms show no fairness improvement in cases where the unfairness is due primarily to topology: i.e., cases where traditional unfairness is high, but Stratified Disparity is low. On **Bowdoin**, because the protected attribute is not a primary driver of disparity, all algorithms exhibit comparable performance under stratified evaluation. In particular, methods optimized solely for predictive accuracy and those explicitly designed for fairness achieve similar levels of Stratified Disparity, indicating that algorithmic design choices regarding fairness play a limited role in this network.

In contrast, clear algorithmic differences emerge on the **Polblogs** network. For instance, CrossWalk, which explicitly leverages attribute information to encourage random walks between nodes with different protected attribute, performs worse than the plain DeepWalk baseline. Conversely, FLIP, which suppresses attribute information, achieves the lowest Stratified Disparity. These results highlight that algorithmic treatment of attribute signals becomes a critical factor in fairness outcomes. Similarly, the fairness-oriented FairWalk method does not effectively address stratified fairness concerns. This sort of analysis is useful when selecting a link prediction method: for example, much like in other machine learning workflows, one could perform a fairness analysis on a subset of the graph, and then use the results to select an algorithm for the larger dataset.

Table 2. Elbow Point of Stratified Disparity Curve

Dataset	Jaccard	DeepWalk	FairLP	UGE	EDITS	FairWalk	CrossWalk	FLIP
Dutch	0.0613	0.0187	0.0749	-0.0034	0.0518	0.0972	0.2509	0.0913
Bowdoin	-0.0086	-0.0083	-0.0099	-0.0020	0.0090	-0.0045	-0.0047	0.0001
DM	0.0116	0.0212	0.0089	0.0056	0.0016	0.0226	0.0092	0.0093
Graphics	-0.0048	0.0197	-0.0064	-0.0028	-0.0074	-0.0118	-0.0149	-0.0090
Pokec	-0.0010	-0.0010	0.0003	0.0009	-0.0027	-0.0003	0.0019	-0.0008
Polblogs	0.0041	0.0513	0.0146	0.0117	0.0071	0.0451	0.0894	-0.0078
UNC	0.0099	0.0005	0.0077	0.0040	0.0027	-0.0006	0.0004	0.0024
Recid.	0.0155	0.0014	0.0049	-0.0845	0.0150	-0.0018	-0.0001	0.0035

Our results suggest that *Stratified Disparity* can give more nuanced fairness insights than existing aggregate fairness metrics. In some networks (such as Bowdoin), a large portion of unfairness appears due to differences in node degree, and once those differences are accounted for, much of the unfairness is explained. This does not suggest that the unfairness was not real, but rather provides an explanation for it, allowing practitioners to design targeted mitigation strategies rather than applying global fairness corrections that may be ineffective or misleading. In other networks (such as Polblogs), the unfairness was not due primarily to differences in degree. In this network, both protected groups were well-represented among high-degree nodes. In this type of network, other causes of unfairness should be explored.

6 Conclusion

In this work, we introduced *Stratified Disparity*, a fairness metric for evaluating link prediction algorithms on networks. Unlike traditional fairness metrics, which aggregate individuals in a protected group together, *Stratified Disparity* further subdivides nodes based on degree, thus exploring the extent to which link prediction unfairness is due to differences in degree. Through experiments on several real-world datasets, using eight popular link prediction algorithms, we demonstrated that *Stratified Disparity* provides more nuanced insights into the origins of unfairness than aggregating together all individuals in a protected group. In future work, we will explore stratification by other topological properties and categorize networks based on their behavior with respect to this metric.

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